

Technical Document #389: Cloud Computing - Comprehensive Overview

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Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Kubernetes orchestrates containerized applications across clusters of machines. It handles deployment, scaling, and management of containerized workloads.

Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. AWS EC2 and Azure VMs are popular IaaS offerings.

Platform as a Service (PaaS) provides a platform for developing and deploying applications. Heroku and Google App Engine are PaaS examples.

Containerization packages applications with their dependencies. Docker creates lightweight, portable containers that run consistently across environments.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files.
- Serverless computing allows developers to run code without managing servers.
- Microservices architecture structures applications as a collection of loosely coupled services.